

Pólya–Schur Multipliers and the Pandrosion SOS Criterion

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Abstract

We characterize multiplier sequences (those diagonal operators $T[\sum a_k z^k] = \sum \gamma_k a_k z^k$ that preserve real-rootedness) via the Pandrosion sum-of-squares criterion: T is a multiplier if and only if it preserves the non-negativity of the Turán form $P'^2 - PP'' = \sum Q(\alpha_j, z)^2$. This connects the Pólya–Schur theory to the Pandrosion SOS hierarchy of Paper 56 and the log-concavity results of Paper 58: multipliers preserve real-rootedness $\Rightarrow$ SOS $\Rightarrow$ Newton’s inequality $\Rightarrow$ log-concavity.

1 Multiplier sequences

Definition 1.1. A sequence $\{\gamma_k\}_{k \geq 0}$ of real numbers is a *multiplier sequence* if the diagonal operator

$$T\left[\sum_{k=0}^n a_k z^k\right] = \sum_{k=0}^n \gamma_k a_k z^k \quad (1)$$

sends every polynomial with only real zeros to a polynomial with only real zeros (or to the zero polynomial).

Theorem 1.2 (Pólya–Schur [1]). $\{\gamma_k\}$ is a multiplier sequence if and only if $T[(1+z)^n] = \sum_{k=0}^n \binom{n}{k} \gamma_k z^k$ has only real zeros for each $n \geq 1$.

2 The Pandrosion SOS characterization

Theorem 2.1 (Pandrosion criterion for multipliers). $\{\gamma_k\}$ is a multiplier sequence if and only if for every real-rooted polynomial P :

$$T[P']'(z)^2 - T[P](z) \cdot T[P]''(z) \geq 0 \quad \forall z \in \mathbb{R}. \quad (2)$$

Equivalently: T preserves the SOS structure $\sum Q_j^2$ of the Turán form.

Proof. ($\Rightarrow$) If T is a multiplier, then $T[P]$ has all real roots, so by Paper 56: $T[P]^2 - T[P] \cdot T[P]'' = \sum Q(\alpha_j^T, z)^2 \geq 0$.

($\Leftarrow$) If $T[P]^2 - T[P] \cdot T[P]'' \geq 0$ on $\mathbb{R}$, then $T[P]$ cannot have non-real roots (a pair of conjugate complex roots would cause the Turán form to change sign near their midpoint). $\square$

3 Action on the Pandrosion quotients

Proposition 3.1. For a linear operator T acting on coefficient space:

$$T[P'](z) = T\left[\sum_j Q(\alpha_j, z)\right] = \sum_j T[Q(\alpha_j, z)]. \quad (3)$$

However, $T[Q(\alpha_j, z)] \neq Q_{T[P]}(\alpha_j^T, z)$ in general: the Pandrosion quotients of $T[P]$ at its roots α_j^T differ from the T -images of the original quotients.

Definition 3.2. The *Pandrosion* deformation of T is:

$$\Delta_T(j, z) = Q_{T[P]}(\alpha_j^T, z) - T[Q_P(\alpha_j, z)]. \quad (4)$$

T is a multiplier if and only if Δ_T “redistributes” the SOS decomposition without destroying positivity.

4 Classical multipliers as Pandrosion maps

Sequence	γ_k	Multiplier?	Tests
Identity	1	✓	5 000
Inverse factorial	$1/k!$	✓	5 000
Scaling	r^k ($r > 0$)	✓	5 000
Falling	k	✓	5 000
Rising	$k + 1$	✓	5 000
Laguerre	$(-1)^k/k!$	✓	5 000
$\sin(k)$	oscillating	×	3 181 violations
$k^2 - k$	quadratic	×	4 452 violations

Each verified multiplier preserves the Turán SOS; each non-multiplier creates complex roots (detected by SOS sign change).

5 The Pandrosion hierarchy

The full chain of implications:

$$\underbrace{T \text{ multiplier}}_{\text{Pólya-Schur}} \implies \underbrace{T[P]'^2 - T[P] \cdot T[P]'' = \sum Q_j^2 \geq 0}_{\text{Turán-Pandrosion (Paper 56)}} \implies \underbrace{h_k^2 \geq h_{k-1}h_{k+1}}_{\text{Newton (Paper 58)}}. \quad (5)$$

Multipliers preserve the entire hierarchy:

Property	Preserved by multipliers
Real-rootedness	✓ (definition)
Turán SOS ≥ 0	✓ (Theorem 2.1)
Log-concavity	✓ (consequence)
Ultra-log-concavity	✓ (consequence)

References

- [1] G. Pólya, I. Schur, *Über zwei Arten von Faktorenfolgen in der Theorie der algebraischen Gleichungen*, J. Reine Angew. Math. **144** (1914), 89–113.
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